

Sunil Kumar Narayanan

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🌐 My Webpage

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Education

- 2022–2024 **Master of Science in Computer Science**, *Georgia Institute of Technology*.
Advisor: Dr. Lu Gan
GPA: 4.0/4.0
- 2017–2021 **Bachelor of Technology in Software Engineering**, *SRM Institute of Science and Technology*.
GPA: 80.2%

Publications

- 2026 Sunil Kumar Narayanan, Lingjun Zhao, Lu Gan, and Yongsheng Chen. Diffusion-Denoised Hyperspectral Gaussian Splatting. In *Proceedings of the International Conference on 3D Vision (3DV)*, 2026. To appear. Also available as arXiv:2505.21890.
- 2025 Gerry Chen, Sunil Kumar Narayanan, Thomas Gautier Ottou, Benjamin Missaoui, Harsh Muriki, Cédric Pradalier, and Yongsheng Chen. Hyperspectral Neural Radiance Fields. In *CVPR Workshop on Neural Fields Beyond Conventional Cameras*, 2025. Spotlight Presentation. Also available as arXiv:2403.14839.

Patents & Intellectual Property

- 2025 Sunil Kumar Narayanan, Jie Liu, and Yanli Liu. Image Quality Detection Method and Related Products Thereof. U.S. Patent Application US 19/303,458, filed August 19, 2025. Patent pending, 2025.

Research & Professional Experience

- Aug 2023 – **Graduate Student Researcher**, *Georgia Institute of Technology*, Atlanta, GA.
- Aug 2025
- Co-developed a novel Hyperspectral 3D Reconstruction method **HS-NeRF** leveraging Neural Radiance Fields (NeRFs) with wavelength-conditioned rendering, achieving **robust spectral accuracy** for non-RGB novel view synthesis tasks. Enabled novel use cases in biology and recycling with enhanced wavelength-specific visualization; **Accepted to CVPR 2025: Neural Fields Beyond Conventional Cameras Workshop for a spotlight presentation**.
 - Developed **DD-HGS**, a framework combining 3D Gaussian Splatting with diffusion modeling to mitigate sensor noise, achieving a 20 dB gain in PSNR and competitive rendering speeds; **accepted to 3DV 2026**.
 - Developed synthetic data generation pipelines using neural feature-map conditioned ControlNets for controllable adverse-weather simulation (e.g., rain, fog), enhancing spatial and semantic consistency, enabling robust perception models.
 - Developed a LiDAR-based digital twin covering 20 km of the Atlanta Beltline using 3D reconstruction and semantic understanding techniques, enabling scalable simulation and analysis for urban planning and autonomous vehicle navigation.
- Jun 2024 – **AI Research Intern**, *Shein AI*, Palo Alto, CA.
- Aug 2024
- Worked on 3D Virtual Try-on for the Shein Studio platform.
 - Designed a multi-stage filtering system to detect excessively retouched product images using smoothness detection and geometric consistency checks.
 - Enhanced low-quality image detection accuracy by 25% and reduced manual review overhead; this system resulted in a **U.S. patent application (US 19/303,458, filed Aug 19, 2025)**.

- Aug 2022 – **Senior Research Scientist**, *Fastcode AI*, Bengaluru, India.
- Feb 2023
 - Led a 500-person annotation team to build the first large-scale Indic fashion dataset; designed an entropy-based quality control framework that improved label consistency by 90%.
 - Designed object detection and tracking models optimized for UAV-based autonomous drones, achieving 25% faster inference and 20% higher accuracy.
- Sep 2021 – **Computer Vision Engineer**, *VisionBox.ai*, Remote.
- Aug 2022
 - Developed deep learning models for disease and insect detection in agricultural drone imagery, improving field-level diagnostics.
 - Built analytics tools from sports video footage to extract game-specific features and player metrics.
 - Implemented image stitching and retrieval pipelines to extract spatial insights from high-resolution aerial datasets.
- Jul 2021 – **Computer Vision Intern**, *Sony Research India*, Bengaluru, India.
- Dec 2021
 - Researched and developed a novel cross-modal attention mechanism to model the interplay between vision, audio, and text features.
 - The resulting model surpassed state-of-the-art benchmarks and was successfully integrated into internal Sony platforms.

Fellowships & Awards

- 2020 Awarded **Bronze Medal** in the **SIIC Melanoma Detection Challenge** (Kaggle Competition) for achieving an F1 score of 0.938 in skin lesion classification.

Skills

Technical Skills Python, C++, Bash, PyTorch, TensorFlow, Hugging Face, Diffusers, OpenCV, 3D Gaussian Splatting, NeRF, PyTorch3D, COLMAP, ControlNet, Object Detection, Segmentation, Multimodal Learning, Docker, Git, GCP, Linux, LaTeX

Peer Review

- 2025, 2026 **Reviewer**, International Conference on 3D Vision (3DV)

Teaching Assistantship

- Aug 2023 – **Teaching Assistant**, **OMSCS Academic Integrity**, *Georgia Institute of Technology*, Atlanta, GA.
- May 2024